



COMPETENCY STANDARD

FOR

SEMICONDUCTOR TESTING ENGINEERING

(SEMICONDUCTOR SECTOR)

01 September 2026

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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The Competency Standard for Semiconductor Testing Engineering serves as a foundational document for the development of curricula, teaching and learning materials, practical lab activities and assessment tools specifically tailored for the Semiconductor sector. This document ensures that training activities align with industry requirements, enabling individuals who successfully meet the established standards through assessment to become qualified professionals in relevant entry-level job roles within the sector.

Developed under the Skills for Industry Competitiveness and Innovation Program (SICIP) and implemented by the Bangladesh Semiconductor Industry Association (BSIA), this document addresses the skills requirements critical for Semiconductor Testing Engineering. It is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

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INTRODUCTION

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The semiconductor testing function ensures that integrated circuits meet functional, electrical, quality and reliability requirements before they are released to customers. Semiconductor test engineers work across characterization, wafer sort, final test, production support, yield analysis, hardware interface and automated test program development. The Semiconductor Testing Engineering course aims to develop a skilled workforce capable of performing these functions in accordance with industry requirements. The course covers semiconductor foundations, test and measurement fundamentals, automated test equipment and the Advantest V93000/SmartTest environment, test program development and pattern engineering, DFT concepts, digital, analog and mixed-signal testing methodologies, test hardware and interface design, manufacturing test operations, yield engineering, emerging applications and capstone project execution. The course is designed for EEE, ECE, CSE, Applied Physics and relevant engineering graduates who want to build career readiness for job roles such as Semiconductor Test Engineer, ATE Test Engineer, Product/Test Engineer, Test Development Engineer, IC Test Engineer, Yield Engineer and Failure Analysis Support Engineer. The training will be imparted by the Bangladesh Semiconductor Industry Association (BSIA) with funding support from SICIP.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The Competency Standard also suggests integration of YouTube or similar digital platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

The document details the format, sequencing, wording, and layout of the Competency Standard for the occupation which comprises Units of Competency and their corresponding Elements.

OVERVIEW

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies.
- encourages the development and delivery of flexible training that suits individual and industry requirements.
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

The Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at a training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Identification and validation of units of competency and elements for the occupation were made by experts from the semiconductor design, manufacturing and test industry in industry consultative workshops.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competency:

- An overview of all Units of Competency for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:

Occupation Specific Competencies (240 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
BSIA-SEM-TEST-01-O	Explain Semiconductor Industry and IC Testing Fundamentals	1. Describe semiconductor industry, business models and IC lifecycle 2. Interpret wafer fabrication, assembly, packaging and test flow 3. Explain the importance of testing, quality and reliability 4. Identify roles and career paths in semiconductor testing	08
BSIA-SEM-TEST-02-O	Apply Electronics and Semiconductor Device Fundamentals	1. Explain basic electronics and semiconductor physics 2. Interpret digital, analog and mixed-signal circuits 3. Identify memory and power devices 4. Interpret device specifications and datasheets	12
BSIA-SEM-TEST-03-O	Perform Test and Measurement Fundamentals	1. Use electrical measurement principles 2. Apply instrumentation basics 3. Perform parametric measurement and characterization 4. Ensure measurement accuracy and signal integrity	16
BSIA-SEM-TEST-04-O	Operate Automated Test Equipment Fundamentals	1. Describe ATE architecture and tester resources 2. Identify test cell components 3. Apply handler, prober and loadboard concepts 4. Observe tester safety and production requirements	12

Code	Unit of Competency	Elements of Competency	Duration (Hours)
BSIA-SEM-TEST-05-O	Use Advantest V93000 and SmarTest Environment	1. Set up V93000 project environment 2. Configure pins, resources and device setup 3. Develop and organize test program structure 4. Use debugging tools in SmarTest	20
BSIA-SEM-TEST-06-O	Develop Test Programs and Pattern Engineering	1. Create test flows and specifications 2. Develop vectors and patterns 3. Set up timing and levels 4. Execute, debug and optimize patterns	28
BSIA-SEM-TEST-07-O	Apply Design for Testability Concepts	1. Explain DFT fundamentals 2. Interpret scan architecture and ATPG overview 3. Describe JTAG, boundary scan and MBIST concepts 4. Relate design features to production test	16
BSIA-SEM-TEST-08-O	Perform Digital IC Testing Methodologies	1. Apply functional and structural digital testing 2. Interpret scan and boundary scan test methods 3. Use memory testing concepts and fault models 4. Optimize digital production testing	12
BSIA-SEM-TEST-09-O	Perform Analog Testing Fundamentals	1. Conduct DC parametric testing 2. Perform AC and analog performance measurements 3. Analyze analog data and limits 4. Debug analog test failures	28

Code	Unit of Competency	Elements of Competency	Duration (Hours)
BSIA-SEM-TEST-10-O	Perform Mixed-Signal Testing Fundamentals	1. Explain ADC and DAC concepts 2. Configure mixed-signal measurements 3. Analyze conversion and performance data 4. Debug mixed-signal test issues	28
BSIA-SEM-TEST-11-O	Apply Test Hardware and Interface Design	1. Explain loadboard and fixture fundamentals 2. Apply PCB considerations for test 3. Plan power delivery, grounding and signal routing 4. Verify test hardware readiness	24
BSIA-SEM-TEST-12-O	Apply Manufacturing Test Operations and Yield Engineering	1. Explain wafer sort and final test operations 2. Apply multisite testing and binning concepts 3. Analyze yield and characterization data 4. Support failure analysis and production optimization	12
BSIA-SEM-TEST-13-O	Recognize Emerging Trends and Industry Applications in Testing	1. Identify testing needs for automotive, AI/HPC and IoT devices 2. Describe advanced packaging and RF testing overview 3. Recognize future technologies and industry trends 4. Relate trends to test engineering skills	08
BSIA-SEM-TEST-14-O	Complete Capstone Project and Practical Assessment	1. Plan end-to-end test project 2. Develop and execute test program tasks	16

Code	Unit of Competency	Elements of Competency	Duration (Hours)
		3. Debug, analyze and report test results 4. Present project outcomes and complete assessment	
Total Hours			240 Hrs.

COMPETENCY STANDARD: SEMICONDUCTOR TESTING ENGINEERING

Occupation Specific Competencies

Unit of Competency: EXPLAIN SEMICONDUCTOR INDUSTRY AND IC TESTING FUNDAMENTALS	Nominal Duration: 08 hrs.	Unit Code: BSIA-SEM-TEST-01-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to understand the semiconductor industry, IC lifecycle and the role of testing in ensuring quality and reliability of semiconductor products. It specifically includes the tasks of describing the semiconductor industry, business models and IC lifecycle, interpreting wafer fabrication, assembly, packaging and test flow, explaining the importance of testing, quality and reliability and identifying roles and career paths in semiconductor testing.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Describe semiconductor industry, business models and IC lifecycle	1.1. <u>Semiconductor industry structure and business models</u> are described according to industry practice. 1.2. Roles of fabless companies, foundries, IDMs and OSATs in the semiconductor value chain are differentiated. 1.3. <u>IC lifecycle stages</u> from specification to production and field deployment are interpreted correctly. 1.4. Position of testing within the overall IC lifecycle is explained. 1.5. Major semiconductor market segments and product categories are identified.
2. Interpret wafer fabrication, assembly, packaging and test flow	2.1. <u>Wafer fabrication process steps</u> are outlined at overview level according to standard manufacturing flow. 2.2. <u>Assembly and packaging processes</u> are described in relation to device protection and interconnection. 2.3. <u>Testing stages</u> across the manufacturing flow are identified and sequenced correctly. 2.4. Wafer sort, package test and final test stages are differentiated according to their purpose. 2.5. Interfaces between fabrication, assembly, packaging and test operations are explained.

Elements of Competency	Performance Criteria
3. Explain the importance of testing, quality and reliability	3.1.Relationships among test cost, quality, yield and time-to-market are explained. 3.2.<u>Quality and reliability requirements</u> of semiconductor products are described according to industry expectations. 3.3.Production screening and outgoing quality objectives are interpreted correctly. 3.4.<u>Reliability terminology</u> is interpreted and applied in context. 3.5.Consequences of test escapes and field failures are explained with examples.
4. Identify roles and career paths in semiconductor testing	4.1.<u>Job roles in semiconductor test engineering</u> are identified according to industry organization structures. 4.2.Responsibilities and deliverables of each testing role are described. 4.3.Required technical and workplace skills for testing roles are identified. 4.4.<u>Workplace documentation practices</u> used in test engineering are described. 4.5.Career progression pathways in semiconductor testing are summarized.

Range of Variables

Variable	Range May include but not limited to:
1. Semiconductor industry structure and business models	1.1.Fabless Company 1.2.Foundry 1.3.Integrated Device Manufacturer (IDM) 1.4.Outsourced Semiconductor Assembly and Test (OSAT) 1.5.Design Service Provider 1.6.EDA and Equipment Vendors
2. IC lifecycle stages	2.1.Specification and Design 2.2.Wafer Fabrication 2.3.Assembly and Packaging 2.4.Wafer Sort 2.5.Final Test 2.6.Qualification and Field Deployment

Variable	Range May include but not limited to:
3. Wafer fabrication process steps	3.1.Oxidation and Deposition 3.2.Photolithography 3.3.Etching 3.4.Ion Implantation and Diffusion 3.5.Metallization and Interconnect 3.6.Wafer Probe Preparation
4. Assembly and packaging processes	4.1.Wafer Dicing 4.2.Die Attach 4.3.Wire Bonding / Flip-Chip Bonding 4.4.Encapsulation and Molding 4.5.Lead/Ball Attach and Marking 4.6.Package Types (QFP, QFN, BGA, CSP)
5. Testing stages	5.1.Characterization 5.2.Design Validation 5.3.Wafer Sort 5.4.Package/Final Test 5.5.Production Testing 5.6.Reliability and Qualification Testing
6. Quality and reliability requirements	6.1.Outgoing Quality Level (DPPM) 6.2.Yield Targets 6.3.Reliability Screening 6.4.Burn-in Requirements 6.5.Customer Quality Standards 6.6.Automotive/Industrial Grade Requirements
7. Reliability terminology	7.1.Failure Rate 7.2.Infant Mortality 7.3.Bathtub Curve 7.4.Mean Time to Failure (MTTF) 7.5.Burn-in 7.6.Qualification Stress Tests
8. Job roles in semiconductor test engineering	8.1.Test Engineer 8.2.ATE Test Engineer 8.3.Product Engineer

Variable	Range May include but not limited to:
	8.4. Test Development Engineer 8.5. Yield Engineer 8.6. Failure Analysis Support Engineer
9. Workplace documentation practices	9.1. Test Plans 9.2. Test Reports 9.3. Specification Documents 9.4. Engineering Change Notes 9.5. Logbooks and Checklists 9.6. Data Summary Reports

Curricular Content Guide:

1. Underpinning Knowledge	1. Semiconductor ecosystem, value chain and business models (fabless, foundry, IDM, OSAT). 2. IC lifecycle from specification, design and fabrication to assembly, test and deployment. 3. Overview of wafer fabrication, assembly and packaging processes. 4. Testing stages: characterization, validation, wafer sort, final test and reliability testing. 5. Relationship among test cost, quality, yield and time-to-market. 6. Quality, reliability and outgoing quality concepts in semiconductor manufacturing. 7. Job roles, responsibilities and career pathways in semiconductor test engineering. 8. Workplace documentation and reporting practices in test engineering.
2. Underpinning Skills	1. Interpreting IC lifecycle and semiconductor manufacturing flow diagrams 2. Differentiating wafer sort, package test and final test stages 3. Identifying testing stages, deliverables and interfaces in the manufacturing flow 4. Explaining quality, reliability and yield relationships with examples 5. Identifying semiconductor testing job roles and required skills 6. Preparing short technical summaries of industry and test flow topics

	7. Communicating test engineering roles and responsibilities clearly
3. Underpinning Attitudes	1. Commitment to occupational health and safety 2. Promptness in carrying out activities 3. Sincerity and honesty to duties 4. Environmental concerns 5. Eagerness to learn 6. Tidiness and timeliness 7. Respect for rights of peers and seniors in the workplace 8. Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 1. Classroom or simulated workplace 2. Semiconductor process and manufacturing flow charts 3. IC lifecycle and value chain diagrams 4. Sample datasheets and test flow diagrams 5. Industry case studies and presentation materials 6. Multimedia projector and learning materials

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1. described semiconductor industry, business models and IC lifecycle 2. interpreted wafer fabrication, assembly, packaging and test flow 3. explained the importance of testing, quality and reliability 4. identified roles and career paths in semiconductor testing
2. Methods of Assessment	Methods of assessment should include: 1. Written test 2. Practical demonstration 3. Oral questioning 4. Portfolio (optional)
3. Context of Assessment	1. Competency assessment must be done in an assessment/training center or in an actual or simulated workplace after completion of the training. 2. Assessment should be done by a certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY ELECTRONICS AND SEMICONDUCTOR DEVICE FUNDAMENTALS	Nominal Duration: 12 hrs.	Unit Code: BSIA-SEM-TEST- 02-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply electronics and semiconductor device fundamentals for test engineering activities. It specifically includes the tasks of explaining basic electronics and semiconductor physics, interpreting digital, analog and mixed-signal circuits, identifying memory and power devices and interpreting device specifications and datasheets.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Explain basic electronics and semiconductor physics	1.1. <u>Basic circuit quantities and laws</u> are explained according to electronics fundamentals. 1.2. Semiconductor material properties and PN junction behavior are described correctly. 1.3. <u>Semiconductor devices</u> and their functions are described according to device physics. 1.4. Device operating regions are identified from characteristic curves and specifications. 1.5. Device behavior relevant to testing requirements is interpreted correctly.
2. Interpret digital, analog and mixed-signal circuits	2.1. <u>Digital logic circuits and functions</u> are interpreted according to circuit fundamentals. 2.2. <u>Analog circuit functions</u> are interpreted in relation to signal processing behavior. 2.3. Mixed-signal device functions combining digital and analog domains are explained. 2.4. <u>Common device pins and interfaces</u> including supplies, clocks and references are identified. 2.5. Functional blocks are mapped to likely test requirements.
3. Identify memory and power devices	3.1. <u>Memory device types</u> and their key characteristics are described correctly. 3.2. Memory organization, addressing and key parameters are interpreted. 3.3. <u>Power devices</u> and their ratings are identified according to device categories.

Elements of Competency	Performance Criteria
	3.4.Safe operating requirements of power devices are interpreted from specifications. 3.5.Testing considerations for memory and power devices are identified.
4. Interpret device specifications and datasheets	4.1.<u>Datasheet sections and parameters</u> are located and explained according to standard datasheet structure. 4.2.Absolute maximum ratings and recommended operating conditions are identified and differentiated. 4.3.<u>Electrical characteristics</u> are interpreted with correct units and conditions. 4.4.Test limits are derived from device specifications according to requirements. 4.5.Specification ambiguities and missing information are identified and clarified.

Range of Variables

Variable	Range May include but not limited to:
1. Basic circuit quantities and laws	1.1.Voltage, Current and Resistance 1.2.Ohm's Law 1.3.Kirchhoff's Laws 1.4.Power and Energy 1.5.Capacitance and Inductance 1.6.Frequency and Time Period
2. Semiconductor devices	2.1.Diodes 2.2.Bipolar Junction Transistors (BJTs) 2.3.MOSFETs 2.4.CMOS Logic Devices 2.5.Operational Amplifiers 2.6.Voltage Regulators
3. Digital logic circuits and functions	3.1.Logic Gates 3.2.Flip-Flops and Registers 3.3.Counters and Shift Registers 3.4.Multiplexers and Decoders 3.5.Digital I/O Buffers

Variable	Range May include but not limited to:
	3.6.Clock and Reset Circuits
4. Analog circuit functions	4.1.Amplification 4.2.Filtering 4.3.Oscillation 4.4.Voltage Reference 4.5.Comparators 4.6.Signal Conditioning
5. Common device pins and interfaces	5.1.Power Supply Pins (VDD/VSS) 5.2.Ground Pins 5.3.Clock Inputs 5.4.Reset Pins 5.5.Reference Pins 5.6.I/O and Interface Pins
6. Memory device types	6.1.SRAM 6.2.DRAM 6.3.Flash Memory 6.4.EEPROM 6.5.ROM 6.6.Emerging Memory (Overview)
7. Power devices	7.1.Power MOSFETs 7.2.Power Diodes 7.3.Voltage Regulators (LDO, DC-DC) 7.4.Power Management ICs 7.5.IGBTs (Overview) 7.6.Protection Devices
8. Datasheet sections and parameters	8.1.Feature Summary and Pin Description 8.2.Absolute Maximum Ratings 8.3.Recommended Operating Conditions 8.4.Electrical Characteristics 8.5.Timing Diagrams 8.6.Application Information
9. Electrical characteristics	9.1.Supply Voltage and Current 9.2.Input/Output Voltage Levels

Variable	Range May include but not limited to:
	9.3. Leakage Current 9.4. Timing Parameters 9.5. Gain and Offset 9.6. Conversion Accuracy

Curricular Content Guide:

1. Underpinning Knowledge	1. Basic electronics: circuit quantities, laws and passive components. 2. Semiconductor physics fundamentals: materials, PN junction, diode, BJT and MOSFET operation. 3. Digital, analog and mixed-signal circuit fundamentals. 4. Memory device types, organization and key parameters. 5. Power device categories, ratings and safe operating areas. 6. Datasheet structure, specification types and electrical parameters. 7. Derivation of test limits from device specifications. 8. Relationship between device behavior and testing requirements.
2. Underpinning Skills	1. Explaining device operation and operating regions from characteristics 2. Reading and interpreting datasheets accurately 3. Identifying device pins, supplies, clocks and interfaces 4. Interpreting digital, analog and mixed-signal circuit functions 5. Identifying memory and power device parameters 6. Deriving simple test limits from specifications 7. Relating device behavior to testing needs
3. Underpinning Attitudes	1. Commitment to occupational health and safety 2. Promptness in carrying out activities 3. Sincerity and honesty to duties 4. Environmental concerns 5. Eagerness to learn 6. Tidiness and timeliness 7. Respect for rights of peers and seniors in the workplace 8. Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 1. Classroom/computer laboratory

	2. Sample IC datasheets 3. Basic circuit diagrams and device characteristic curves 4. Calculator and worksheets 5. Device specification examples 6. Reference books and learning materials
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Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1. explained basic electronics and semiconductor physics 2. interpreted digital, analog and mixed-signal circuits 3. identified memory and power devices 4. interpreted device specifications and datasheets
2. Methods of Assessment	Methods of assessment should include: 1. Written test 2. Practical demonstration 3. Oral questioning 4. Portfolio (optional)
3. Context of Assessment	1. Competency assessment must be done in an assessment/training center or in an actual or simulated workplace after completion of the training. 2. Assessment should be done by a certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM TEST AND MEASUREMENT FUNDAMENTALS	Nominal Duration: 16 hrs.	Unit Code: BSIA-SEM-TEST-03-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform basic electrical measurements, use instruments and apply measurement accuracy principles in semiconductor testing. It specifically includes the tasks of using electrical measurement principles, applying instrumentation basics, performing parametric measurement and characterization and ensuring measurement accuracy and signal integrity.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Use electrical measurement principles	1.1. <u>Electrical measurements</u> and their principles are explained according to measurement fundamentals. 1.2. Measurement ranges and limits are selected according to device and test requirements. 1.3. Measurement methods for voltage, current, resistance, frequency and timing are applied correctly. 1.4. <u>Measurement safety practices</u> are followed during all measurement activities. 1.5. Measurement results are recorded with correct units, resolution and conditions.
2. Apply instrumentation basics	2.1. <u>Test and measurement instruments</u> are identified and their functions explained. 2.2. Instrument connections are made according to safe measurement configuration. 2.3. <u>Instrument settings</u> are configured according to measurement requirements. 2.4. Instrument readings are captured and recorded correctly. 2.5. Instrument care, handling and basic calibration checks are performed according to workplace procedures.
3. Perform parametric measurement and characterization	3.1. <u>Parametric tests</u> are performed as per specified test conditions. 3.2. Force and measure principles are applied for parametric measurements. 3.3. <u>Characterization data</u> are collected and organized systematically.

Elements of Competency	Performance Criteria
	3.4.Measurement results are compared with specification limits. 3.5.Deviations and out-of-limit results are flagged and reported.
4. Ensure measurement accuracy and signal integrity	4.1.<u>Measurement accuracy concepts</u> including resolution, calibration and uncertainty are interpreted. 4.2.Repeatability and reproducibility of measurements are evaluated. 4.3.<u>Signal integrity issues and noise sources</u> are identified in measurement setups. 4.4.Grounding, cabling and probing practices are applied to reduce measurement errors. 4.5.Corrective actions are applied and verified to improve measurement quality.

Range of Variables

Variable	Range May include but not limited to:
1. Electrical measurements	1.1.Voltage Measurement 1.2.Current Measurement 1.3.Resistance Measurement 1.4.Leakage Measurement 1.5.Frequency and Duty Cycle Measurement 1.6.Timing Measurement
2. Measurement safety practices	2.1.Voltage and Current Limit Setting 2.2.Correct Probe Rating 2.3.ESD-safe Handling 2.4.Power Sequencing 2.5.Instrument Grounding 2.6.Safe Connection/Disconnection
3. Test and measurement instruments	3.1.Digital Multimeter (DMM) 3.2.Oscilloscope 3.3.DC Power Supply 3.4.Function/Signal Generator 3.5.Logic Analyzer 3.6.Source Measure Unit (SMU) Overview
4. Instrument settings	4.1.Range and Resolution

Variable	Range May include but not limited to:
	4.2.Coupling (AC/DC) 4.3.Trigger Settings 4.4.Sampling Rate and Timebase 4.5.Probe Attenuation 4.6.Averaging and Filtering
5. Parametric tests	5.1.Continuity Test 5.2.Leakage Current Test 5.3.Supply Current Measurement 5.4.Output Voltage Levels (VOH/VOL) 5.5.Input Thresholds (VIH/VIL) 5.6.Resistance Measurement
6. Characterization data	6.1.Measured Values vs Limits 6.2.Data Tables and Logs 6.3.Trend and Sweep Data 6.4.Statistical Summaries (Min/Max/Mean) 6.5.Distribution Data 6.6.Test Condition Records
7. Measurement accuracy concepts	7.1.Accuracy 7.2.Resolution 7.3.Calibration 7.4.Measurement Uncertainty 7.5.Repeatability 7.6.Traceability
8. Signal integrity issues and noise sources	8.1.Ground Loops 8.2.Cable and Probe Loading 8.3.Electromagnetic Interference 8.4.Thermal Drift 8.5.Contact Resistance 8.6.Power Supply Noise

Curricular Content Guide:

1. Underpinning Knowledge	1. Electrical measurement principles for voltage, current, resistance, frequency and timing.
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	2. Operation and configuration of DMMs, oscilloscopes, power supplies and signal sources. 3. Force and measure concepts for parametric testing. 4. Parametric measurement and characterization procedures. 5. Measurement accuracy, resolution, calibration and uncertainty concepts. 6. Signal integrity fundamentals: noise, grounding, loading and probing. 7. Data recording, organization and comparison with limits. 8. Safety practices in electrical measurement activities.
2. Underpinning Skills	1. Connecting instruments safely and correctly 2. Configuring instrument settings for required measurements 3. Measuring electrical quantities accurately 4. Performing parametric tests according to test conditions 5. Recording, organizing and analyzing measurement data 6. Identifying and reducing noise and signal integrity issues 7. Troubleshooting measurement setup errors
3. Underpinning Attitudes	1. Commitment to occupational health and safety 2. Promptness in carrying out activities 3. Sincerity and honesty to duties 4. Environmental concerns 5. Eagerness to learn 6. Tidiness and timeliness 7. Respect for rights of peers and seniors in the workplace 8. Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 1. Electronics/measurement laboratory 2. DMMs, oscilloscopes, power supplies and signal generators 3. Test boards or training kits 4. Probes, cables and connection accessories 5. Measurement worksheets and data log templates 6. Sample device specifications with test limits

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1. used electrical measurement principles 2. applied instrumentation basics 3. performed parametric measurement and characterization 4. ensured measurement accuracy and signal integrity
2. Methods of Assessment	Methods of assessment should include: 1. Written test 2. Practical demonstration 3. Oral questioning 4. Portfolio (optional)
3. Context of Assessment	1. Competency assessment must be done in an assessment/training center or in an actual or simulated workplace after completion of the training. 2. Assessment should be done by a certified assessor or occupation-specific industry expert.

Unit of Competency: OPERATE AUTOMATED TEST EQUIPMENT FUNDAMENTALS	Nominal Duration: 12 hrs.	Unit Code: BSIA-SEM-TEST-04-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to understand automated test equipment architecture, tester resources, test cells and production test requirements. It specifically includes the tasks of describing ATE architecture and tester resources, identifying test cell components, applying handler, prober and loadboard concepts and observing tester safety and production requirements.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Describe ATE architecture and tester resources	1.1. <u>ATE architecture and major subsystems</u> are described according to tester documentation. 1.2. <u>Tester resources</u> for digital, analog, power and timing functions are identified. 1.3. Resource capabilities and specifications are interpreted from tester documentation. 1.4. Tester resource limitations and sharing constraints are interpreted correctly. 1.5. Device test requirements are mapped to appropriate tester resources.
2. Identify test cell components	2.1. <u>Test cell components</u> are identified and their roles explained. 2.2. DUT board, loadboard and probe card functions are differentiated correctly. 2.3. Handler and prober functions in the test cell are described. 2.4. <u>Test cell interfaces</u> between tester, hardware and handling equipment are identified. 2.5. Test cell configuration for wafer sort and final test is differentiated.
3. Apply handler, prober and loadboard concepts	3.1. <u>Handler operations and configurations</u> are described according to production test practice. 3.2. <u>Prober operations</u> including wafer handling and alignment are described at overview level. 3.3. Handler and prober interactions with the tester are interpreted correctly. 3.4. Loadboard and probe card interface requirements are interpreted from documentation.

Elements of Competency	Performance Criteria
	3.5.Docking, alignment and contact considerations are explained.
4. Observe tester safety and production requirements	4.1.<u>Tester safety practices</u> for electrical and mechanical hazards are followed. 4.2.ESD control requirements in the test environment are observed at all times. 4.3.<u>Production test requirements</u> including lot handling and traceability are described. 4.4.Production test data collection requirements are interpreted correctly. 4.5.Abnormal tester conditions are recognized and reported according to procedures.

Range of Variables

Variable	Range May include but not limited to:
1. ATE architecture and major subsystems	1.1.Test Head 1.2.Tester Mainframe 1.3.Workstation/Controller 1.4.Pin Electronics 1.5.Power Supplies (DPS) 1.6.Clock and Timing System
2. Tester resources	2.1.Digital I/O Channels 2.2.Parametric Measurement Unit (PMU) 2.3.Device Power Supply (DPS) 2.4.Arbitrary Waveform Generator (AWG) 2.5.Digitizer 2.6.Timing and Memory Resources
3. Test cell components	3.1.Tester 3.2.Handler 3.3.Prober 3.4.Manipulator 3.5.Loadboard/DUT Board 3.6.Probe Card
4. Test cell interfaces	4.1.Tester-Loadboard Interface 4.2.Loadboard-Socket Interface

Variable	Range May include but not limited to:
	4.3.Probe Card-Wafer Interface 4.4.Handler Docking Interface 4.5.GPIB/LAN Communication 4.6.Utility Connections (Air, Vacuum, Power)
5. Handler operations and configurations	5.1.Pick and Place Handling 5.2.Gravity Feed Handling 5.3.Turret Handling (Overview) 5.4.Temperature Setpoint Control 5.5.Binning and Sorting 5.6.Jam Recovery
6. Prober operations	6.1.Wafer Loading 6.2.Wafer Alignment 6.3.Probe Contact (Touchdown) 6.4.Stepping and Indexing 6.5.Probe Card Setup 6.6.Wafer Mapping
7. Tester safety practices	7.1.Electrical Hazard Awareness 7.2.Mechanical Motion Safety 7.3.ESD Control 7.4.High Current Protection 7.5.Emergency Stop Usage 7.6.Safe Docking Procedures
8. Production test requirements	8.1.Lot Tracking 8.2.Wafer Sort Flow 8.3.Final Test Flow 8.4.Multisite Testing 8.5.Binning Requirements 8.6.Test Data Logging (STDF Overview)

Curricular Content Guide:

1. Underpinning Knowledge	1. ATE architecture, subsystems and tester resource types. 2. Digital, analog, power and timing resources and their capabilities.
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	3. Test cell components: tester, handler, prober, loadboard and probe card. 4. Handler and prober operations and interfaces with the tester. 5. Loadboard and probe card interface requirements. 6. Production test environment: wafer sort, final test, multisite and binning. 7. Test data collection and lot traceability requirements. 8. ATE safety, ESD control and safe working practices.
2. Underpinning Skills	1. Identifying ATE subsystems and tester resources 2. Mapping DUT requirements to tester resources 3. Identifying test cell components and their interfaces 4. Interpreting handler, prober and loadboard configurations 5. Interpreting production test flow and data requirements 6. Following tester safety and ESD procedures 7. Recognizing and reporting abnormal tester conditions
3. Underpinning Attitudes	1. Commitment to occupational health and safety 2. Promptness in carrying out activities 3. Sincerity and honesty to duties 4. Environmental concerns 5. Eagerness to learn 6. Tidiness and timeliness 7. Respect for rights of peers and seniors in the workplace 8. Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 1. ATE block diagrams and tester documentation 2. Sample loadboard and probe card illustrations 3. Test cell layout diagrams 4. ESD and safety guidelines 5. Production test flow examples 6. Videos/simulations of handler and prober operation

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1. described ATE architecture and tester resources 2. identified test cell components
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	3. applied handler, prober and loadboard concepts 4. observed tester safety and production requirements
2. Methods of Assessment	Methods of assessment should include: 1. Written test 2. Practical demonstration 3. Oral questioning 4. Portfolio (optional)
3. Context of Assessment	1. Competency assessment must be done in an assessment/training center or in an actual or simulated workplace after completion of the training. 2. Assessment should be done by a certified assessor or occupation-specific industry expert.

Unit of Competency: USE ADVANTEST V93000 AND SMARTEST ENVIRONMENT	Nominal Duration: 20 hrs.	Unit Code: BSIA-SEM-TEST-05-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to use the Advantest V93000 platform and SmarTest environment for semiconductor test program setup and debugging. It specifically includes the tasks of setting up the V93000 project environment, configuring pins, resources and device setup, developing and organizing test program structure and using debugging tools in SmarTest.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Set up V93000 project environment	1.1. <u>V93000 platform architecture and SmarTest framework</u> are explained according to platform documentation. 1.2. Project workspace and environment are created according to standard procedures. 1.3. <u>Project files and directories</u> are organized according to workplace standards. 1.4. Required libraries, device files and settings are configured correctly. 1.5. Environment setup is verified before test program development.
2. Configure pins, resources and device setup	2.1. <u>Pin configuration items</u> are defined according to device pin map. 2.2. Tester resource allocation is performed according to device test requirements. 2.3. <u>Device setup parameters</u> including levels and timing references are configured. 2.4. Channel assignments and resource mappings are verified against hardware documentation. 2.5. Configuration files are reviewed, validated and corrected as required.
3. Develop and organize test program structure	3.1. <u>Test program components</u> are created according to SmarTest program structure. 3.2. Test flow and test suites are organized in logical sequence. 3.3. <u>Test methods and parameters</u> are linked to test suites as required.

Elements of Competency	Performance Criteria
	3.4.Specifications, levels, timing and patterns are associated with the test program correctly. 3.5.Program organization and naming follow workplace standards.
4. Use debugging tools in SmarTest	4.1.Compilation and runtime errors are identified from messages and logs. 4.2.SmarTest debugging tools are used to analyze test execution. 4.3.Test results and datalog outputs are interpreted correctly. 4.4.Debug actions are applied to isolate and resolve setup and execution issues. 4.5.Corrective actions are implemented, verified and documented.

Range of Variables

Variable	Range May include but not limited to:
1. V93000 platform architecture and SmarTest framework	1.1.Test Head and Card Cage 1.2.Pin Electronics Cards 1.3.Device Power Supplies 1.4.SmarTest Software Environment 1.5.Workstation and Operating Environment 1.6.Offline/Simulation Mode
2. Project files and directories	2.1.Device Directory Structure 2.2.Pin Configuration Files 2.3.Levels and Timing Files 2.4.Pattern Directories 2.5.Test Flow Files 2.6.Test Method Libraries
3. Pin configuration items	3.1.Pin Names and Groups 3.2.Channel Assignments 3.3.Pin Types (I/O, Power, Utility) 3.4.Site Assignments 3.5.Resource Types 3.6.Pin Attributes
4. Device setup parameters	4.1.Supply Voltages 4.2.Input/Output Levels

Variable	Range May include but not limited to:
	4.3.Termination Settings 4.4.Clock and Timing References 4.5.Test Conditions 4.6.Temperature Conditions (Overview)
5. Test program components	5.1.Test Flow 5.2.Test Suites 5.3.Test Methods 5.4.Specification Sets 5.5.Level Sets 5.6.Timing Sets and Patterns
6. Test methods and parameters	6.1.Continuity Test Method 6.2.Leakage Test Method 6.3.Functional Test Method 6.4.DC/Parametric Test Methods 6.5.Test Method Parameters 6.6.Limit Settings
7. SmarTest debugging tools	7.1.Message/Report Windows 7.2.Datalog Viewer 7.3.Pattern Debug Tools 7.4.Waveform/Shmoo Tools 7.5.Breakpoints and Step Execution 7.6.Result Analysis Windows
8. Debug actions	8.1.Reviewing Error Logs 8.2.Checking Pin and Level Setup 8.3.Verifying Pattern and Timing 8.4.Re-running Test Suites 8.5.Isolating Failing Tests 8.6.Documenting Corrections

Curricular Content Guide:

1. Underpinning Knowledge	1. V93000 platform architecture and SmarTest software framework. 2. Project workspace creation and file organization. 3. Pin configuration, channel assignment and resource allocation.
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	4. Device setup: levels, timing references and test conditions. 5. SmarTest test program structure: flows, suites, methods and specifications. 6. Linking specifications, levels, timing and patterns in a test program. 7. Debugging workflow: error identification, analysis and resolution. 8. Workplace standards for program organization and documentation.
2. Underpinning Skills	1. Creating and configuring V93000/SmarTest test projects 2. Configuring pins, resources and device setup 3. Organizing test program files and structures 4. Developing and linking test flows, suites and methods 5. Using debugging tools to analyze test execution 6. Interpreting datalogs and result windows 7. Resolving and documenting setup and execution problems
3. Underpinning Attitudes	1. Commitment to occupational health and safety 2. Promptness in carrying out activities 3. Sincerity and honesty to duties 4. Environmental concerns 5. Eagerness to learn 6. Tidiness and timeliness 7. Respect for rights of peers and seniors in the workplace 8. Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 1. V93000/SmarTest environment or offline simulator 2. Computer laboratory 3. Sample device specifications and pin maps 4. Training test program examples 5. Platform documentation and quick reference guides 6. Debug logs and datalog examples

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1. set up V93000 project environment 2. configured pins, resources and device setup
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	3. developed and organized test program structure 4. used debugging tools in SmarTest
2. Methods of Assessment	Methods of assessment should include: 1. Written test 2. Practical demonstration 3. Oral questioning 4. Portfolio (optional)
3. Context of Assessment	1. Competency assessment must be done in an assessment/training center or in an actual or simulated workplace after completion of the training. 2. Assessment should be done by a certified assessor or occupation-specific industry expert.

Unit of Competency: DEVELOP TEST PROGRAMS AND PATTERN ENGINEERING	Nominal Duration: 28 hrs.	Unit Code: BSIA-SEM-TEST-06-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to develop semiconductor test programs, generate patterns, configure timing and levels and debug pattern execution. It specifically includes the tasks of creating test flows and specifications, developing vectors and patterns, setting up timing and levels and executing, debugging and optimizing patterns.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Create test flows and specifications	1.1. <u>Test flow concepts and structures</u> are interpreted according to test methodology. 1.2. <u>Test items</u> are defined with limits, units and conditions according to device requirements. 1.3. Test sequencing and dependencies are established in logical order. 1.4. Binning assignments are created according to test flow requirements. 1.5. <u>Specification sets</u> are created and managed according to device requirements.
2. Develop vectors and patterns	2.1. <u>Vector and pattern requirements</u> are identified from device functional behavior. 2.2. Patterns are developed or imported according to test objectives. 2.3. <u>Pattern file formats</u> and conversion requirements are interpreted correctly. 2.4. Pattern naming conventions and organization are followed according to workplace standards. 2.5. Pattern syntax and structure are verified before execution.
3. Set up timing and levels	3.1. <u>Timing setup items</u> including timing sets and edge placement are configured according to device specifications. 3.2. Waveform formats are selected according to signal requirements. 3.3. <u>Level setup items</u> including drive and compare levels are established from datasheet values.

Elements of Competency	Performance Criteria
	3.4.Setup and hold requirements are applied to timing configuration. 3.5.Timing and level setups are verified before pattern execution.
4. Execute, debug and optimize patterns	4.1.Patterns are executed and results are reviewed systematically. 4.2.<u>Pattern failures</u> are analyzed using logs and debug tools. 4.3.<u>Debug techniques</u> are applied to isolate failure causes. 4.4.Timing, level and pattern corrections are implemented and re-verified. 4.5.Test execution is optimized for robustness, test time and efficiency.

Range of Variables

Variable	Range May include but not limited to:
1. Test flow concepts and structures	1.1.Test Flow Sequence 1.2.Test Suites and Groups 1.3.Conditional Branching 1.4.Stop-on-Fail / Continue-on-Fail 1.5.Binning Structure 1.6.Flow Control Statements
2. Test items	2.1.Continuity Test 2.2.Leakage Test 2.3.Functional Test 2.4.Scan Test 2.5.Analog/Parametric Test 2.6.Mixed-Signal Test
3. Specification sets	3.1.Test Limits (Min/Max) 3.2.Voltage and Current Conditions 3.3.Timing Conditions 3.4.Variables and Equations 3.5.Temperature Conditions 3.6.Guardband Values
4. Vector and pattern requirements	4.1.Functional Patterns 4.2.Scan Patterns

Variable	Range May include but not limited to:
	4.3.Memory Patterns 4.4.Initialization Sequences 4.5.Loop and Repeat Structures 4.6.Pattern Labels and Subroutines
5. Pattern file formats	5.1.STIL 5.2.WGL 5.3.VCD/EVCD 5.4.ATP/Native Tester Format 5.5.Pattern Compiler Outputs 5.6.Pattern Conversion Logs
6. Timing setup items	6.1.Timing Sets (Tsets) 6.2.Period and Frequency 6.3.Edge Placement (Drive/Compare Edges) 6.4.Waveform Formats (NRZ, RZ, SBC) 6.5.Setup and Hold Margins 6.6.Clock Definitions
7. Level setup items	7.1.VIH/VIL (Drive Levels) 7.2.VOH/VOL (Compare Levels) 7.3.Termination Voltage 7.4.Power Supply Levels 7.5.Current Clamp Settings 7.6.Reference Levels
8. Pattern failures	8.1.Miscompare Failures 8.2.Timeout Errors 8.3.Setup/Configuration Errors 8.4.Resource Conflicts 8.5.Syntax Errors 8.6.Intermittent/Marginal Failures
9. Debug techniques	9.1.Failure Log Analysis 9.2.Waveform Inspection 9.3.Shmoo Plotting 9.4.Pattern Single-Stepping 9.5.Margin Analysis

Variable	Range May include but not limited to:
	9.6. Loop-on-Fail Execution

Curricular Content Guide:

1. Underpinning Knowledge	1. Test flow architecture, sequencing and binning concepts. 2. Test specification management: limits, conditions and variables. 3. Vector and pattern engineering fundamentals and file formats. 4. Timing setup: timing sets, edge placement and waveform formats. 5. Level setup: drive, compare, termination and supply levels. 6. Pattern execution and result interpretation. 7. Pattern debugging techniques: logs, waveforms and shmoo analysis. 8. Test time optimization and robustness considerations.
2. Underpinning Skills	1. Creating and configuring test flows and binning 2. Defining test items with limits and conditions 3. Developing, importing and organizing patterns 4. Configuring timing sets, edges and waveform formats 5. Establishing level setups from device specifications 6. Executing patterns and debugging failures systematically 7. Optimizing test execution for time and robustness
3. Underpinning Attitudes	1. Commitment to occupational health and safety 2. Promptness in carrying out activities 3. Sincerity and honesty to duties 4. Environmental concerns 5. Eagerness to learn 6. Tidiness and timeliness 7. Respect for rights of peers and seniors in the workplace 8. Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 1. ATE software environment or simulator 2. Sample patterns and test flows 3. Device specification and pin map 4. Pattern format references and conversion tools 5. Debug logs and waveform examples

	6. Test program development guidelines
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Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1. created test flows and specifications 2. developed vectors and patterns 3. set up timing and levels 4. executed, debugged and optimized patterns
2. Methods of Assessment	Methods of assessment should include: 1. Written test 2. Practical demonstration 3. Oral questioning 4. Portfolio (optional)
3. Context of Assessment	1. Competency assessment must be done in an assessment/training center or in an actual or simulated workplace after completion of the training. 2. Assessment should be done by a certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY DESIGN FOR TESTABILITY CONCEPTS	Nominal Duration: 16 hrs.	Unit Code: BSIA-SEM-TEST-07-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply DFT concepts and understand the relationship between chip design and semiconductor production testing. It specifically includes the tasks of explaining DFT fundamentals, interpreting scan architecture and ATPG overview, describing JTAG, boundary scan and MBIST concepts and relating design features to production test.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Explain DFT fundamentals	1.1. Need for DFT in modern ASIC and SoC testing is explained with justification. 1.2. <u>Faults, defects and failures</u> are differentiated according to test terminology. 1.3. <u>Fault models</u> used in structural testing are described correctly. 1.4. Test coverage and fault coverage concepts are interpreted correctly. 1.5. <u>DFT terminology</u> is interpreted and used correctly in context.
2. Interpret scan architecture and ATPG overview	2.1. <u>Scan design elements</u> and scan chain architecture are described according to DFT methodology. 2.2. Scan shift and capture operations are explained correctly. 2.3. <u>ATPG concepts</u> including purpose and basic flow are explained. 2.4. Scan compression concept and benefits are recognized. 2.5. Scan pattern application on ATE is interpreted at overview level.
3. Describe JTAG, boundary scan and MBIST concepts	3.1. <u>JTAG/IEEE 1149.1 elements</u> are described according to the standard architecture. 3.2. Boundary scan applications for interconnect and infrastructure testing are described. 3.3. <u>Memory BIST elements</u> and operation are explained at concept level. 3.4. Logic BIST concept is recognized at overview level. 3.5. DFT features are linked to expected test patterns and test modes.

Elements of Competency	Performance Criteria
4. Relate design features to production test	4.1. <u>DFT outputs used in ATE test programs</u> are identified correctly. 4.2. Test modes and test access mechanisms are interpreted from design documentation. 4.3. Design-test interactions and constraints are interpreted correctly. 4.4. <u>Design and test collaboration items</u> are identified for effective test bring-up. 4.5. Impact of DFT quality on production test cost and coverage is explained.

Range of Variables

Variable	Range May include but not limited to:
1. Faults, defects and failures	1.1. Physical Defects 1.2. Manufacturing Process Variations 1.3. Stuck-at Faults 1.4. Timing-related Faults 1.5. Functional Failures 1.6. Parametric Failures
2. Fault models	2.1. Stuck-at Fault Model 2.2. Transition Delay Fault Model 2.3. Path Delay Fault Model (Overview) 2.4. Bridging Fault Model 2.5. Cell-aware Faults (Overview) 2.6. Memory Fault Models
3. DFT terminology	3.1. Controllability and Observability 3.2. Test Coverage 3.3. Fault Coverage 3.4. Test Point 3.5. Test Mode 3.6. Test Access Mechanism
4. Scan design elements	4.1. Scan Flip-Flops 4.2. Scan Chains 4.3. Scan Enable Signal

Variable	Range May include but not limited to:
	4.4.Scan In/Scan Out Ports 4.5.Shift and Capture Cycles 4.6.Scan Compression Logic
5. ATPG concepts	5.1Automatic Test Pattern Generation Flow 5.2.Fault List Generation 5.3.Pattern Generation and Compaction 5.4.Fault Simulation 5.5.Coverage Reports 5.6.Pattern Formats for ATE
6. JTAG/IEEE 1149.1 elements	6.1.Test Access Port (TAP) 6.2.TAP Controller 6.3.Instruction Register 6.4.Boundary Scan Register 6.5.Bypass Register 6.6.JTAG Instructions (EXTEST, SAMPLE, BYPASS)
7. Memory BIST elements	7.1.BIST Controller 7.2.Address Generator 7.3.Data/Pattern Generator 7.4.Response Comparator 7.5.March Algorithms 7.6.Repair Support (Overview)
8. DFT outputs used in ATE test programs	8.1.Scan Patterns (STIL/WGL) 8.2.Test Protocol Files 8.3.Pin Constraint Files 8.4.Test Mode Setup Sequences 8.5.Coverage Reports 8.6.Pattern Application Notes
9. Design and test collaboration items	9.1.Pin Muxing and Test Modes 9.2.Clocking for Test 9.3.Power Constraints during Test 9.4.Pattern Timing Requirements 9.5.Diagnosis Support 9.6.ECO Impact on Patterns

Curricular Content Guide:

1. Underpinning Knowledge	1. Purpose and benefits of Design for Testability in ASIC/SoC development. 2. Faults, defects, failures and fault model fundamentals. 3. Scan design architecture: scan chains, shift and capture operations. 4. ATPG flow, coverage metrics and scan compression overview. 5. JTAG/IEEE 1149.1 architecture and boundary scan applications. 6. Memory BIST architecture and march algorithm concepts. 7. DFT outputs and their application in ATE test programs. 8. Design-test interactions, constraints and collaboration practices.
2. Underpinning Skills	1. Interpreting DFT documentation and terminology 2. Describing scan operation, ATPG flow and coverage reports 3. Explaining JTAG, boundary scan and MBIST operation 4. Relating scan patterns to test program needs 5. Identifying test modes, access mechanisms and constraints 6. Linking DFT features to production test requirements 7. Communicating effectively with design and DFT teams
3. Underpinning Attitudes	1. Commitment to occupational health and safety 2. Promptness in carrying out activities 3. Sincerity and honesty to duties 4. Environmental concerns 5. Eagerness to learn 6. Tidiness and timeliness 7. Respect for rights of peers and seniors in the workplace 8. Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 1. DFT architecture diagrams and examples 2. Sample scan pattern files (STIL/WGL) 3. JTAG and MBIST overview materials 4. ATPG coverage report samples 5. ATE/DFT interface examples 6. Reference standards documentation (IEEE 1149.1 overview)

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1. explained DFT fundamentals 2. interpreted scan architecture and ATPG overview 3. described JTAG, boundary scan and MBIST concepts 4. related design features to production test
2. Methods of Assessment	Methods of assessment should include: 1. Written test 2. Practical demonstration 3. Oral questioning 4. Portfolio (optional)
3. Context of Assessment	1. Competency assessment must be done in an assessment/training center or in an actual or simulated workplace after completion of the training. 2. Assessment should be done by a certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM DIGITAL IC TESTING METHODOLOGIES	Nominal Duration: 12 hrs.	Unit Code: BSIA-SEM-TEST-08-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform digital IC testing methodologies including functional, structural, scan, boundary scan and memory test concepts. It specifically includes the tasks of applying functional and structural digital testing, interpreting scan and boundary scan test methods, using memory testing concepts and fault models and optimizing digital production testing.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Apply functional and structural digital testing	1.1. <u>Digital test types</u> are differentiated according to test methodology. 1.2. Functional test objectives and coverage characteristics are explained. 1.3. <u>Digital test conditions</u> are configured according to device specification. 1.4. Digital tests are executed and pass/fail results are interpreted correctly. 1.5. Functional and structural test trade-offs are explained.
2. Interpret scan and boundary scan test methods	2.1. <u>Scan test execution items</u> are described according to scan methodology. 2.2. Scan pattern pass/fail behavior is interpreted from tester results. 2.3. <u>Boundary scan test applications</u> are explained according to JTAG methodology. 2.4. Digital patterns are linked to expected device responses. 2.5. Scan-related failures are categorized for further analysis.
3. Use memory testing concepts and fault models	3.1. <u>Memory test algorithms</u> are explained according to memory testing methodology. 3.2. <u>Memory fault types</u> are described according to memory fault models. 3.3. Memory test patterns and address/data behaviors are interpreted. 3.4. Memory test results and fail signatures are interpreted correctly.

Elements of Competency	Performance Criteria
	3.5.Memory repair and redundancy concepts are recognized at overview level.
4. Optimize digital production testing	4.1.Test time contributors are identified and analyzed. 4.2.<u>Test time and coverage trade-offs</u> are analyzed according to production requirements. 4.3.<u>Production test optimization strategies</u> are considered and applied where appropriate. 4.4.Multisite testing implications for digital tests are interpreted. 4.5.Optimization results are verified without compromising test quality.

Range of Variables

Variable	Range May include but not limited to:
1. Digital test types	1.1.Continuity Test 1.2.Leakage Test 1.3.Functional Test 1.4.Structural (Scan) Test 1.5.Boundary Scan Test 1.6.At-speed Test (Overview)
2. Digital test conditions	2.1.Supply Voltage Corners 2.2.Input/Output Levels 2.3.Test Frequency 2.4.Timing Margins 2.5.Temperature Conditions (Overview) 2.6.Test Modes
3. Scan test execution items	3.1.Scan Mode Entry 3.2.Scan Shift Operation 3.3.Capture Cycle 3.4.Scan Unload and Compare 3.5.Scan Compression Operation 3.6.Scan Fail Logging
4. Boundary scan test applications	4.1.Interconnect Testing 4.2.Infrastructure Testing

Variable	Range May include but not limited to:
	4.3.EXTEST/SAMPLE Instructions 4.4.Chain Integrity Test 4.5.Cluster Testing (Overview) 4.6.In-System Access
5. Memory test algorithms	5.1.March C- 5.2.March Algorithms (General) 5.3.Checkerboard Pattern 5.4.Address Walking Pattern 5.5.Data Retention Test 5.6.GalPat (Overview)
6. Memory fault types	6.1.Stuck-at Faults 6.2.Transition Faults 6.3.Coupling Faults 6.4.Address Decoder Faults 6.5.Data Retention Faults 6.6.Neighborhood Pattern Sensitive Faults (Overview)
7. Test time and coverage trade-offs	7.1.Pattern Count Reduction 7.2.Test Reordering 7.3.Early Fail Detection 7.4.Coverage vs Test Time Analysis 7.5.Sampling Strategies 7.6.Guardband Adjustment
8. Production test optimization strategies	8.1.Multisite Testing 8.2.Concurrent Testing 8.3.Test Elimination/Merging 8.4.Binning Optimization 8.5.Adaptive Test (Overview) 8.6.Test Flow Restructuring

Curricular Content Guide:

1. Underpinning Knowledge	1. Digital IC testing methodologies: functional, structural and parametric. 2. Scan test execution: shift, capture and compare operations.
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	3. Boundary scan and JTAG-based digital test methods. 4. Memory test algorithms, fault models and fail signatures. 5. Digital test condition configuration and corner testing. 6. Pass/fail interpretation and failure categorization. 7. Test time analysis and optimization strategies. 8. Multisite and production efficiency considerations.
2. Underpinning Skills	1. Configuring digital test conditions from specifications 2. Running functional, scan and boundary scan tests 3. Interpreting pass/fail results and fail logs 4. Applying memory test algorithms and interpreting results 5. Analyzing miscompares and categorizing failures 6. Analyzing test time and coverage trade-offs 7. Improving digital test flow efficiency
3. Underpinning Attitudes	1. Commitment to occupational health and safety 2. Promptness in carrying out activities 3. Sincerity and honesty to duties 4. Environmental concerns 5. Eagerness to learn 6. Tidiness and timeliness 7. Respect for rights of peers and seniors in the workplace 8. Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 1. Digital test program examples 2. Patterns and expected response files 3. ATE environment or simulator 4. Memory test algorithm references 5. Failure log and datalog examples 6. Test time analysis worksheets

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1. applied functional and structural digital testing 2. interpreted scan and boundary scan test methods 3. used memory testing concepts and fault models
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	4. optimized digital production testing
2. Methods of Assessment	Methods of assessment should include: 1. Written test 2. Practical demonstration 3. Oral questioning 4. Portfolio (optional)
3. Context of Assessment	1. Competency assessment must be done in an assessment/training center or in an actual or simulated workplace after completion of the training. 2. Assessment should be done by a certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM ANALOG TESTING FUNDAMENTALS	Nominal Duration: 28 hrs.	Unit Code: BSIA-SEM-TEST-09-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to conduct analog semiconductor tests including DC parametric, AC and analog performance measurements. It specifically includes the tasks of conducting DC parametric testing, performing AC and analog performance measurements, analyzing analog data and limits and debugging analog test failures.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Conduct DC parametric testing	1.1. <u>DC parametric tests</u> and their requirements are interpreted from device specifications. 1.2. <u>Tester analog resources</u> are selected and configured for DC measurements. 1.3. Force and measure conditions are set according to test requirements. 1.4. DC measurements are executed and results are recorded accurately. 1.5. DC measurement results are evaluated against specification limits.
2. Perform AC and analog performance measurements	2.1. <u>AC and analog performance tests</u> are identified from device requirements. 2.2. Stimulus signals are configured according to measurement requirements. 2.3. <u>Analog measurement setups</u> including capture and analysis settings are configured. 2.4. AC and analog measurements are executed as applicable. 2.5. Analog performance results are compared with specifications correctly.
3. Analyze analog data and limits	3.1. Analog test data are organized and reviewed systematically. 3.2. <u>Statistical analysis items</u> are applied to analog measurement data. 3.3. Outliers and marginal results are identified from distributions. 3.4. <u>Limit setting considerations</u> including guardbands are discussed and applied.

Elements of Competency	Performance Criteria
	3.5.Data-driven recommendations are prepared for limit or test improvements.
4. Debug analog test failures	4.1.<u>Analog failure causes</u> are categorized as hardware, setup or device-related. 4.2.Instrument and tester resource settings are checked systematically. 4.3.<u>Analog debug techniques</u> are applied to isolate failure causes. 4.4.Bench correlation is performed where applicable to verify tester measurements. 4.5.Corrective actions are implemented, verified and documented.

Range of Variables

Variable	Range May include but not limited to:
1. DC parametric tests	1.1.Continuity Test 1.2.Leakage Current (IIL/IIH/IOZ) 1.3.Supply Current (IDD/ICC) 1.4.Output Levels (VOH/VOL) 1.5.Input Thresholds (VIH/VIL) 1.6.On-Resistance Measurement
2. Tester analog resources	2.1.Parametric Measurement Unit (PMU) 2.2.Device Power Supply (DPS) 2.3.Arbitrary Waveform Generator (AWG) 2.4.Digitizer 2.5.Analog Pin Units 2.6.Time Measurement Unit (Overview)
3. AC and analog performance tests	3.1.Gain Measurement 3.2.Bandwidth Measurement 3.3.Offset Voltage 3.4.Slew Rate 3.5.Noise Measurement (Overview) 3.6.Frequency Response
4. Analog measurement setups	4.1.Stimulus Amplitude and Frequency 4.2.Sampling Rate and Capture Depth

Variable	Range May include but not limited to:
	4.3.Input Ranges 4.4.Filtering and Averaging 4.5.Trigger and Synchronization 4.6.Load Conditions
5. Statistical analysis items	5.1.Mean and Standard Deviation 5.2.Minimum/Maximum Values 5.3.Histograms and Distributions 5.4.Cp/Cpk (Overview) 5.5.Outlier Detection 5.6.Repeatability Analysis
6. Limit setting considerations	6.1.Datasheet Limits 6.2.Guardbands 6.3.Measurement Uncertainty 6.4.Process Variation 6.5.Customer Requirements 6.6.Yield Impact
7. Analog failure causes	7.1.Contact/Continuity Issues 7.2.Instrument Setup Errors 7.3.Loadboard Hardware Faults 7.4.Noise and Interference 7.5.Device Marginality 7.6.Calibration Issues
8. Analog debug techniques	8.1.Signal Path Verification 8.2.Force/Measure Value Checking 8.3.Waveform Capture and Review 8.4.Sweep and Shmoo Analysis 8.5.Bench Correlation 8.6.Substitution Testing

Curricular Content Guide:

1. Underpinning Knowledge	1. DC parametric test principles: continuity, leakage, supply current and levels. 2. ATE analog resources: PMU, DPS, AWG and digitizer operation.
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	3. AC and analog performance measurement methods. 4. Force and measure configuration for analog testing. 5. Statistical analysis of analog test data and distributions. 6. Limit setting, guardbanding and measurement uncertainty considerations. 7. Analog failure categories and debugging methodology. 8. Bench correlation and measurement verification practices.
2. Underpinning Skills	1. Configuring PMU, DPS and analog resources for measurements 2. Setting analog test conditions from specifications 3. Performing DC parametric and AC measurements 4. Organizing and analyzing analog test distributions 5. Identifying outliers and marginal results 6. Applying guardband and limit considerations 7. Debugging analog test setups systematically
3. Underpinning Attitudes	1. Commitment to occupational health and safety 2. Promptness in carrying out activities 3. Sincerity and honesty to duties 4. Environmental concerns 5. Eagerness to learn 6. Tidiness and timeliness 7. Respect for rights of peers and seniors in the workplace 8. Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 1. ATE analog resources or bench instruments 2. Analog test boards or device examples 3. Datasheets with analog test limits 4. Measurement data sets for analysis 5. Statistical analysis tools/spreadsheets 6. Analog debug case examples

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1. conducted DC parametric testing 2. performed AC and analog performance measurements 3. analyzed analog data and limits
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	4. debugged analog test failures
2. Methods of Assessment	Methods of assessment should include: 1. Written test 2. Practical demonstration 3. Oral questioning 4. Portfolio (optional)
3. Context of Assessment	1. Competency assessment must be done in an assessment/training center or in an actual or simulated workplace after completion of the training. 2. Assessment should be done by a certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM MIXED-SIGNAL TESTING FUNDAMENTALS	Nominal Duration: 28 hrs.	Unit Code: BSIA-SEM-TEST-10-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to conduct mixed-signal semiconductor testing including ADC/DAC concepts, measurements and data analysis. It specifically includes the tasks of explaining ADC and DAC concepts, configuring mixed-signal measurements, analyzing conversion and performance data and debugging mixed-signal test issues.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Explain ADC and DAC concepts	1.1. <u>Data converter operating principles</u> are explained according to conversion fundamentals. 1.2. Resolution, sampling and quantization concepts are interpreted correctly. 1.3. <u>Converter architectures</u> are described at overview level. 1.4. Reference, clocking and input range concepts are explained. 1.5. <u>Mixed-signal performance parameters</u> are identified and defined correctly.
2. Configure mixed-signal measurements	2.1. <u>Stimulus and capture requirements</u> are determined from test objectives. 2.2. Signal source and digitizer settings are configured according to measurement requirements. 2.3. <u>Coherent sampling requirements</u> are applied to measurement setup. 2.4. Clocking, reference, timing and level setups are verified before execution. 2.5. Digital and analog resources are coordinated for mixed-signal tests.
3. Analyze conversion and performance data	3.1. Mixed-signal output data are captured and processed correctly. 3.2. <u>Static performance metrics</u> are calculated and interpreted. 3.3. <u>Dynamic performance metrics</u> are calculated and interpreted. 3.4. Analysis methods including histogram and FFT techniques are applied as applicable. 3.5. Measurement results are evaluated against device specifications.

Elements of Competency	Performance Criteria
4. Debug mixed-signal test issues	4.1. Mixed-signal test issues are identified from data signatures and results. 4.2. Timing, noise, reference and data capture problems are isolated systematically. 4.3. Mixed-signal debug tools are used for waveform and data analysis. 4.4. Setup corrections are implemented and measurements re-verified. 4.5. Debug findings and corrective actions are documented.

Range of Variables

Variable	Range May include but not limited to:
1. Data converter operating principles	1.1. Analog-to-Digital Conversion 1.2. Digital-to-Analog Conversion 1.3. Sampling Theorem 1.4. Quantization and Resolution 1.5. Reference Voltage Operation 1.6. Transfer Function
2. Converter architectures	2.1. SAR ADC 2.2. Flash ADC 2.3. Sigma-Delta ADC 2.4. Pipeline ADC (Overview) 2.5. R-2R DAC 2.6. Current Steering DAC (Overview)
3. Mixed-signal performance parameters	3.1. Resolution (Bits) 3.2. INL and DNL 3.3. Offset and Gain Error 3.4. SNR and SINAD 3.5. THD and SFDR (Overview) 3.6. ENOB
4. Stimulus and capture requirements	4.1. Sine Wave Stimulus 4.2. Ramp Stimulus 4.3. Digital Pattern Stimulus 4.4. Capture Depth and Sample Count

Variable	Range May include but not limited to:
	4.5.Sampling Frequency 4.6.Anti-aliasing Considerations
5. Coherent sampling requirements	5.1.Coherency Equation 5.2.Prime Number of Cycles 5.3.Sampling/Signal Frequency Ratio 5.4.Window-free Capture 5.5.Clock Synchronization 5.6.Record Length Selection
6. Static performance metrics	6.1.Offset Error 6.2.Gain Error 6.3.INL 6.4.DNL 6.5.Monotonicity 6.6.Missing Codes
7. Dynamic performance metrics	7.1.SNR 7.2.SINAD 7.3.THD 7.4.SFDR 7.5.ENOB 7.6.Noise Floor
8. Mixed-signal test issues	8.1.Sampling/Coherency Errors 8.2.Clock Jitter Effects 8.3.Reference Instability 8.4.Noise Coupling 8.5.Data Alignment Errors 8.6.Insufficient Settling Time
9. Mixed-signal debug tools	9.1.Waveform Viewers 9.2.FFT/Spectrum Analysis 9.3.Histogram Analysis 9.4.Datalog Review 9.5.Capture Memory Inspection 9.6.Correlation with Bench Data

Curricular Content Guide:

1. Underpinning Knowledge	1. ADC and DAC operating principles and architectures. 2. Sampling theory, quantization, resolution and references. 3. Static performance metrics: offset, gain, INL and DNL. 4. Dynamic performance metrics: SNR, SINAD, THD, SFDR and ENOB. 5. Coherent sampling and capture configuration for mixed-signal test. 6. Histogram and FFT-based analysis techniques. 7. Coordination of digital and analog tester resources. 8. Mixed-signal debugging methodology and tools.
2. Underpinning Skills	1. Configuring stimulus, capture and clocking for mixed-signal tests 2. Applying coherent sampling in measurement setup 3. Capturing and processing conversion data 4. Calculating and interpreting static and dynamic metrics 5. Evaluating results against converter specifications 6. Debugging timing, noise and reference issues 7. Documenting mixed-signal debug findings
3. Underpinning Attitudes	1. Commitment to occupational health and safety 2. Promptness in carrying out activities 3. Sincerity and honesty to duties 4. Environmental concerns 5. Eagerness to learn 6. Tidiness and timeliness 7. Respect for rights of peers and seniors in the workplace 8. Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 1. Mixed-signal test examples and data sets 2. ATE resources or simulation environment 3. Analysis spreadsheets/scripts (FFT, histogram) 4. ADC/DAC datasheets 5. Waveform and spectrum viewing tools 6. Mixed-signal debug case examples

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1. explained ADC and DAC concepts 2. configured mixed-signal measurements 3. analyzed conversion and performance data 4. debugged mixed-signal test issues
2. Methods of Assessment	Methods of assessment should include: 1. Written test 2. Practical demonstration 3. Oral questioning 4. Portfolio (optional)
3. Context of Assessment	1. Competency assessment must be done in an assessment/training center or in an actual or simulated workplace after completion of the training. 2. Assessment should be done by a certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY TEST HARDWARE AND INTERFACE DESIGN	Nominal Duration: 24 hrs.	Unit Code: BSIA-SEM-TEST-11-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to understand test hardware interfaces including loadboards, probe cards, fixtures, PCB considerations, power delivery, grounding and signal routing. It specifically includes the tasks of explaining loadboard and fixture fundamentals, applying PCB considerations for test, planning power delivery, grounding and signal routing and verifying test hardware readiness.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Explain loadboard and fixture fundamentals	1.1. <u>Test hardware components</u> and their functions are explained according to test cell architecture. 1.2. DUT-to-tester interface requirements are identified from device and tester documentation. 1.3. <u>Socket and contactor considerations</u> are described according to device package requirements. 1.4. Probe card types and applications are described at overview level. 1.5. Test hardware constraints and limitations are interpreted correctly.
2. Apply PCB considerations for test	2.1. <u>PCB design considerations for test hardware</u> are described according to design practice. 2.2. Component placement priorities around the DUT are explained. 2.3. <u>Signal integrity considerations</u> are applied at basic level. 2.4. Impedance control requirements for critical signals are recognized. 2.5. Crosstalk and parasitic concerns are identified and mitigation approaches described.
3. Plan power delivery, grounding and signal routing	3.1. <u>Power delivery elements</u> for the DUT are described according to power integrity practice. 3.2. Grounding strategies for mixed-signal test hardware are described correctly. 3.3. <u>Decoupling and filtering requirements</u> are identified according to supply requirements.

Elements of Competency	Performance Criteria
	3.4. Critical signal routing needs are explained according to signal categories. 3.5. Power, ground and signal planning trade-offs are discussed.
4. Verify test hardware readiness	4.1. Hardware verification items are compiled into a readiness checklist. 4.2. Continuity, shorts and connection correctness are checked systematically. 4.3. Hardware verification measurements are performed and recorded. 4.4. Hardware issues are documented and escalated according to procedures. 4.5. Hardware readiness status is reported before test program debug.

Range of Variables

Variable	Range May include but not limited to:
1. Test hardware components	1.1. Loadboard/DUT Board 1.2. Probe Card 1.3. Test Fixture 1.4. Socket and Contactor 1.5. Handler/Prober Interface Hardware 1.6. Interface Pogo Pins/Connectors
2. Socket and contactor considerations	2.1. Package Type Compatibility 2.2. Contact Resistance 2.3. Insertion Count/Lifetime 2.4. Signal Bandwidth 2.5. Thermal Requirements 2.6. Alignment and Placement
3. PCB design considerations for test hardware	3.1. Layer Stack-up 3.2. Trace Width and Spacing 3.3. Via Types and Placement 3.4. Component Placement 3.5. Plane Partitioning 3.6. Material Selection (Overview)

Variable	Range May include but not limited to:
4. Signal integrity considerations	4.1. Controlled Impedance 4.2. Trace Length Matching 4.3. Reflections and Terminations 4.4. Crosstalk Control 4.5. Return Path Continuity 4.6. Parasitic Capacitance/Inductance
5. Power delivery elements	5.1. Power Planes 5.2. Supply Traces and Vias 5.3. Bulk Capacitors 5.4. Decoupling Capacitors 5.5. Ferrite Beads/Filters 5.6. Remote Sense Connections
6. Decoupling and filtering requirements	6.1. Capacitor Value Selection 6.2. Capacitor Placement 6.3. High-frequency Decoupling 6.4. Bulk Decoupling 6.5. Supply Filtering 6.6. Noise Isolation
7. Critical signal routing needs	7.1. High-speed Digital Signals 7.2. Clock Signals 7.3. Analog/Reference Signals 7.4. Differential Pairs 7.5. Power Sense Lines 7.6. Guard Traces/Shielding
8. Hardware verification items	8.1. Continuity Check 8.2. Short Circuit Check 8.3. Visual Inspection 8.4. Component Value Verification 8.5. Interface Alignment Check 8.6. Documentation Cross-check
9. Hardware issues	9.1. Open/Short Circuits 9.2. Wrong Component Values 9.3. Damaged Contacts

Variable	Range May include but not limited to:
	9.4.Alignment Problems 9.5.Intermittent Connections 9.6.Documentation Mismatches

Curricular Content Guide:

1. Underpinning Knowledge	1. Test hardware architecture: loadboards, probe cards, fixtures and sockets. 2. DUT-to-tester interface requirements and constraints. 3. PCB design considerations: stack-up, placement, routing and vias. 4. Signal integrity fundamentals: impedance, crosstalk and return paths. 5. Power delivery, grounding and decoupling strategies. 6. Critical signal routing categories and requirements. 7. Hardware verification methods and readiness checklists. 8. Hardware issue documentation and escalation practices.
2. Underpinning Skills	1. Reading loadboard schematics and layout documentation 2. Identifying DUT interface and socket requirements 3. Applying basic signal integrity considerations 4. Identifying power delivery, grounding and decoupling needs 5. Checking hardware interfaces for continuity and shorts 6. Identifying routing and grounding issues 7. Preparing hardware readiness reports
3. Underpinning Attitudes	1. Commitment to occupational health and safety 2. Promptness in carrying out activities 3. Sincerity and honesty to duties 4. Environmental concerns 5. Eagerness to learn 6. Tidiness and timeliness 7. Respect for rights of peers and seniors in the workplace 8. Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 1. Sample loadboard schematics and layouts 2. PCB layout examples and stack-up diagrams

	3. DMM and continuity check tools 4. Socket and probe card illustrations 5. Hardware checklist templates 6. Signal integrity reference materials
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Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1. explained loadboard and fixture fundamentals 2. applied PCB considerations for test 3. planned power delivery, grounding and signal routing 4. verified test hardware readiness
2. Methods of Assessment	Methods of assessment should include: 1. Written test 2. Practical demonstration 3. Oral questioning 4. Portfolio (optional)
3. Context of Assessment	1. Competency assessment must be done in an assessment/training center or in an actual or simulated workplace after completion of the training. 2. Assessment should be done by a certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY MANUFACTURING TEST OPERATIONS AND YIELD ENGINEERING	Nominal Duration: 12 hrs.	Unit Code: BSIA-SEM-TEST-12-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to support manufacturing test operations, yield analysis, binning, characterization, failure analysis and production optimization. It specifically includes the tasks of explaining wafer sort and final test operations, applying multisite testing and binning concepts, analyzing yield and characterization data and supporting failure analysis and production optimization.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Explain wafer sort and final test operations	1.1. <u>Production test operations</u> are described according to manufacturing test flow. 1.2. Wafer sort and final test flows are differentiated correctly. 1.3. <u>Production test data levels</u> are interpreted according to data hierarchy. 1.4. Production test deliverables are identified according to operational requirements. 1.5. Lot disposition and retest practices are described.
2. Apply multisite testing and binning concepts	2.1. <u>Multisite testing concepts</u> and benefits are explained according to production efficiency requirements. 2.2. Multisite efficiency and site-to-site correlation considerations are interpreted. 2.3. <u>Binning items</u> including soft bins and hard bins are interpreted correctly. 2.4. Bin definitions are reviewed against test flow requirements. 2.5. Binning results are reviewed and used for production actions.
3. Analyze yield and characterization data	3.1. <u>Yield metrics</u> are calculated and summarized correctly. 3.2. <u>Data analysis methods</u> are applied to production and characterization data. 3.3. Parametric distributions and outliers are identified from data. 3.4. Wafer maps and spatial patterns are interpreted at overview level. 3.5. Characterization results are interpreted and summarized for stakeholders.

Elements of Competency	Performance Criteria
4. Support failure analysis and production optimization	4.1. <u>Failure trends and categories</u> are identified and reported from production data. 4.2. Common failure causes are categorized for investigation. 4.3. <u>Production optimization opportunities</u> are identified from test data. 4.4. Inputs for failure analysis requests are prepared according to procedures. 4.5. Technical reports are prepared according to required formats.

Range of Variables

Variable	Range May include but not limited to:
1. Production test operations	1.1. Wafer Sort/Probe 1.2. Package Test 1.3. Final Test 1.4. Characterization 1.5. Qualification Testing 1.6. Quality Assurance Sampling
2. Production test data levels	2.1. Lot-level Data 2.2. Wafer-level Data 2.3. Site-level Data 2.4. Device-level Data 2.5. Test/Parameter-level Data 2.6. Bin Summary Data
3. Multisite testing concepts	3.1. Parallel Site Testing 3.2. Site Count and Efficiency 3.3. Multisite Hardware Requirements 3.4. Site-to-site Correlation 3.5. Resource Sharing Constraints 3.6. Index Time Reduction
4. Binning items	4.1. Soft Bins 4.2. Hard Bins 4.3. Pass Bins 4.4. Fail Bin Categories

Variable	Range May include but not limited to:
	4.5.Quality/Speed Grade Bins 4.6.Retest Bins
5. Yield metrics	5.1.Wafer Sort Yield 5.2.Final Test Yield 5.3.First Pass Yield 5.4.Retest Recovery Rate 5.5.Overall Yield 5.6.Defect Density (Overview)
6. Data analysis methods	6.1.Histograms 6.2.Pareto Analysis 6.3.Trend Charts 6.4.Wafer Maps 6.5.Statistical Summaries 6.6.Correlation Analysis
7. Failure trends and categories	7.1.Parametric Failures 7.2.Functional Failures 7.3.Continuity/Contact Failures 7.4.Site-specific Failures 7.5.Lot-to-lot Variations 7.6.Systematic vs Random Failures
8. Production optimization opportunities	8.1.Test Time Reduction 8.2.Retest Rate Reduction 8.3.Bin Recovery Improvement 8.4.Hardware Maintenance Actions 8.5.Test Limit Optimization 8.6.Process Feedback

Curricular Content Guide:

1. Underpinning Knowledge	1. Manufacturing test operations: wafer sort, package and final test. 2. Production data hierarchy: lot, wafer, site and device level. 3. Multisite testing concepts and efficiency metrics. 4. Binning strategies: soft bins, hard bins and quality grades. 5. Yield calculation, analysis and reporting methods.
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	6. Statistical analysis: histograms, Pareto and trend analysis. 7. Failure trend identification and categorization. 8. Production optimization and failure analysis support practices.
2. Underpinning Skills	1. Interpreting production test flows and deliverables 2. Calculating and summarizing yield metrics 3. Interpreting bin summary reports and wafer maps 4. Analyzing production data with statistical methods 5. Identifying failure trends and outliers 6. Identifying production optimization opportunities 7. Preparing production test and yield reports
3. Underpinning Attitudes	1. Commitment to occupational health and safety 2. Promptness in carrying out activities 3. Sincerity and honesty to duties 4. Environmental concerns 5. Eagerness to learn 6. Tidiness and timeliness 7. Respect for rights of peers and seniors in the workplace 8. Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 1. Sample STDF/CSV production data or reports 2. Data analysis tools/spreadsheets 3. Production flow examples 4. Bin summary and wafer map examples 5. Failure analysis case studies 6. Report templates

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1. explained wafer sort and final test operations 2. applied multisite testing and binning concepts 3. analyzed yield and characterization data 4. supported failure analysis and production optimization
2. Methods of Assessment	Methods of assessment should include: 1. Written test

	2. Practical demonstration 3. Oral questioning 4. Portfolio (optional)
3. Context of Assessment	1. Competency assessment must be done in an assessment/training center or in an actual or simulated workplace after completion of the training. 2. Assessment should be done by a certified assessor or occupation-specific industry expert.

Unit of Competency: RECOGNIZE EMERGING TRENDS AND INDUSTRY APPLICATIONS IN TESTING	Nominal Duration: 08 hrs.	Unit Code: BSIA-SEM-TEST- 13-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to recognize emerging trends and application areas in semiconductor testing. It specifically includes the tasks of identifying testing needs for automotive, AI/HPC and IoT devices, describing advanced packaging and RF testing overview, recognizing future technologies and industry trends and relating trends to test engineering skills.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify testing needs for automotive, AI/HPC and IoT devices	1.1. <u>Application-specific device categories</u> and their characteristics are identified. 1.2. <u>Application-specific test requirements</u> are identified according to end-market demands. 1.3. Reliability and safety expectations for automotive devices are described at overview level. 1.4. Testing challenges for AI/HPC devices are recognized. 1.5. Cost and power considerations for IoT device testing are described.
2. Describe advanced packaging and RF testing overview	2.1. <u>Advanced packaging technologies</u> are recognized at overview level. 2.2. Test challenges introduced by advanced packaging are identified. 2.3. Known good die and interconnect test concepts are described. 2.4. <u>RF testing items</u> and typical measurements are described at overview level. 2.5. RF test equipment categories are recognized.
3. Recognize future technologies and industry trends	3.1. <u>Industry trends and technology drivers</u> are identified from industry information. 3.2. Impacts of trends on test equipment and methods are discussed. 3.3. <u>Data analytics applications in testing</u> are recognized at overview level. 3.4. Sustainability and cost trends in test operations are described.

Elements of Competency	Performance Criteria
	3.5.Sources for continuous industry trend monitoring are identified.
4. Relate trends to test engineering skills	4.1.<u>Skill requirements for emerging application areas</u> are summarized. 4.2.Personal skill gaps are assessed against industry requirements. 4.3.<u>Learning and career development resources</u> are identified. 4.4.Learning and career pathways are prepared according to individual goals. 4.5.Trend awareness is linked to professional development planning.

Range of Variables

Variable	Range May include but not limited to:
1. Application-specific device categories	1.1.Automotive ICs 1.2.AI Accelerators/GPUs 1.3.High-Performance Computing Processors 1.4.IoT Edge Devices 1.5.Power Management ICs 1.6.Sensor and RF Devices
2. Application-specific test requirements	2.1.Zero Defect (DPPM) Targets 2.2.Extended Temperature Testing 2.3.Functional Safety Requirements (Overview) 2.4.High Pin Count/High Power Testing 2.5.Low-cost Test Strategies 2.6.System-Level Test (Overview)
3. Advanced packaging technologies	3.1.2.5D/3D Integration 3.2.Chiplets 3.3.Fan-out Packaging 3.4.Heterogeneous Integration 3.5.Through-Silicon Vias (TSV) 3.6.Wafer-Level Packaging
4. RF testing items	4.1.Output Power Measurement 4.2.Gain and Linearity

Variable	Range May include but not limited to:
	4.3.Frequency Accuracy 4.4.Noise Figure (Overview) 4.5.S-parameters (Overview) 4.6.Modulation Quality (Overview)
5. Industry trends and technology drivers	5.1.AI and Machine Learning Growth 5.2.Automotive Electrification 5.3.5G/6G Communications 5.4.Edge Computing and IoT 5.5.Chiplet Ecosystems 5.6.Test Data Analytics
6. Data analytics applications in testing	6.1.Adaptive Test 6.2.Outlier Detection 6.3.Predictive Maintenance 6.4.Yield Learning 6.5.Real-time Process Feedback 6.6.Machine Learning-based Screening (Overview)
7. Skill requirements for emerging application areas	7.1.ATE Platform Proficiency 7.2.Data Analysis and Scripting 7.3.Test Hardware Knowledge 7.4.DFT Understanding 7.5.Debugging and Problem Solving 7.6.Communication and Reporting
8. Learning and career development resources	8.1.Industry Conferences and Webinars 8.2.Vendor Training Programs 8.3.Technical Journals and Whitepapers 8.4.Online Courses 8.5.Professional Networks 8.6.Internal Mentoring

Curricular Content Guide:

1. Underpinning Knowledge	1. Testing requirements of automotive, AI/HPC and IoT applications. 2. Quality and reliability expectations of safety-critical markets.
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	- Advanced packaging technologies and their test implications. - RF testing fundamentals and typical measurements at overview level. - Semiconductor industry trends and technology drivers. - Data analytics applications in modern test operations. - Skill requirements for emerging test engineering roles. - Career development pathways and learning resources.
2. Underpinning Skills	- Identifying application-specific testing needs - Recognizing advanced packaging test challenges - Describing RF test measurements at overview level - Summarizing industry trends and their impacts - Mapping skills to emerging job requirements - Assessing personal skill gaps - Preparing learning and career development plans
3. Underpinning Attitudes	- Commitment to occupational health and safety - Promptness in carrying out activities - Sincerity and honesty to duties - Environmental concerns - Eagerness to learn - Tidiness and timeliness - Respect for rights of peers and seniors in the workplace - Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: - Industry trend reports and presentations - Application case studies - RF and advanced packaging overview notes - Data analytics application examples - Career pathway templates - Access to online industry resources

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: - identified testing needs for automotive, AI/HPC and IoT devices - described advanced packaging and RF testing overview - recognized future technologies and industry trends
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	4. related trends to test engineering skills
2. Methods of Assessment	Methods of assessment should include: 1. Written test 2. Practical demonstration 3. Oral questioning 4. Portfolio (optional)
3. Context of Assessment	1. Competency assessment must be done in an assessment/training center or in an actual or simulated workplace after completion of the training. 2. Assessment should be done by a certified assessor or occupation-specific industry expert.

Unit of Competency: COMPLETE CAPSTONE PROJECT AND PRACTICAL ASSESSMENT	Nominal Duration: 16 hrs.	Unit Code: BSIA-SEM-TEST-14-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to plan, develop, execute and present an end-to-end semiconductor test project as a capstone activity for practical assessment. It specifically includes the tasks of planning an end-to-end test project, developing and executing test program tasks, debugging, analyzing and reporting test results and presenting project outcomes and completing assessment.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Plan end-to-end test project	1.1. Project objective, device assumptions and test scope are defined clearly. 1.2. <u>Project planning items</u> including test plan and flow are prepared. 1.3. <u>Required resources</u> are identified and organized before project start. 1.4. Tasks are allocated among team members with clear responsibilities. 1.5. Project schedule is prepared and followed according to milestones.
2. Develop and execute test program tasks	2.1. <u>Test program development tasks</u> are implemented as applicable to the project. 2.2. Test flow, patterns, levels, timing and limits are configured according to the plan. 2.3. Test program is executed in simulated or actual environment. 2.4. <u>Execution records</u> are maintained during test runs. 2.5. Results are collected and organized for analysis.
3. Debug, analyze and report test results	3.1. Execution failures are debugged systematically and corrected. 3.2. <u>Data analysis activities</u> are applied to collected test data. 3.3. Findings are validated against project objectives and specifications. 3.4. <u>Technical report components</u> are prepared according to required format. 3.5. Conclusions and recommendations are documented clearly.

Elements of Competency	Performance Criteria
4. Present project outcomes and complete assessment	4.1.<u>Project deliverables</u> are compiled and presented clearly. 4.2.Technical presentation is delivered within allocated time. 4.3.Viva/oral questions are answered with justification. 4.4.<u>Assessment activities</u> are completed according to assessment requirements. 4.5.Feedback is incorporated for final submission.

Range of Variables

Variable	Range May include but not limited to:
1. Project planning items	1.1.Project Objective and Scope 1.2.Test Plan Document 1.3.Test Flow Outline 1.4.Task Breakdown and Schedule 1.5.Risk and Dependency Notes 1.6.Team Role Assignment
2. Required resources	2.1.ATE Software/Simulator Access 2.2.Sample Device Specification 2.3.Computer Workstation 2.4.Data Analysis Tools 2.5.Reference Documentation 2.6.Report/Presentation Templates
3. Test program development tasks	3.1.Pin and Resource Configuration 3.2.Test Flow Creation 3.3.Specification and Limit Setup 3.4.Pattern Integration 3.5.Timing and Level Setup 3.6.Program Validation Runs
4. Execution records	4.1.Run Logs 4.2.Datalogs 4.3.Error and Debug Notes 4.4.Version/Change Records 4.5.Screenshots/Evidence 4.6.Test Condition Records

Variable	Range May include but not limited to:
5. Data analysis activities	5.1.Result Summarization 5.2.Distribution Analysis 5.3.Pass/Fail Analysis 5.4.Outlier Identification 5.5.Correlation Checks 5.6.Yield Summary
6. Technical report components	6.1.Introduction and Objectives 6.2.Methodology 6.3.Test Setup Description 6.4.Results and Analysis 6.5.Debug Summary 6.6.Conclusions and Recommendations
7. Project deliverables	7.1.Test Plan 7.2.Test Program Files 7.3.Debug Report 7.4.Data Analysis Summary 7.5.Technical Report 7.6.Final Presentation
8. Assessment activities	8.1.Practical Demonstration 8.2.Technical Review 8.3.Viva/Oral Questioning 8.4.Report Evaluation 8.5.Peer Review (Where Applicable) 8.6.Final Submission

Curricular Content Guide:

1. Underpinning Knowledge	1. End-to-end semiconductor test program development workflow. 2. Project planning, scoping and resource identification. 3. Test program implementation: flows, patterns, levels, timing and limits. 4. Systematic debugging and data analysis methodology. 5. Technical report structure and documentation standards. 6. Presentation techniques for technical audiences.
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	7. Teamwork, task allocation and schedule management. 8. Assessment requirements and workplace readiness expectations.
2. Underpinning Skills	1. Planning and scoping test projects 2. Developing and executing test flows and programs 3. Maintaining execution records and evidence 4. Debugging failures and analyzing results 5. Preparing technical reports to required formats 6. Delivering technical presentations and answering viva questions 7. Working effectively in project teams
3. Underpinning Attitudes	1. Commitment to occupational health and safety 2. Promptness in carrying out activities 3. Sincerity and honesty to duties 4. Environmental concerns 5. Eagerness to learn 6. Tidiness and timeliness 7. Respect for rights of peers and seniors in the workplace 8. Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 1. ATE environment or simulator 2. Capstone project brief 3. Sample device specification 4. Data analysis tools 5. Report and presentation templates 6. Assessment rubrics and guidelines

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1. planned end-to-end test project 2. developed and executed test program tasks 3. debugged, analyzed and reported test results 4. presented project outcomes and completed assessment
2. Methods of Assessment	Methods of assessment should include: 1. Written test 2. Practical demonstration

	3. Oral questioning 4. Portfolio (optional)
3. Context of Assessment	1. Competency assessment must be done in an assessment/training center or in an actual or simulated workplace after completion of the training. 2. Assessment should be done by a certified assessor or occupation-specific industry expert.

END OF COMPETENCY STANDARD

Workshop/Lab Facility Standard

Course Name:	Semiconductor Testing Engineering
Number of Trainees:	25

Course-wise Training Space:

- Computer Lab: 400 sft
- Test and Measurement Laboratory: 700-800 sft

Major Training Equipment and Training Facilities:

S.N.	Major Training Equipment and Facilities	Required Facilities
1.	Computer/Laptop (Trainer)	01
2.	Printer	01
3.	Multimedia Projector	01
4.	High-performance Desktop/Laptop Workstation	25
5.	ATE Test Program Development Software (Advantest SmarTest Offline/Simulator or equivalent ATE simulation environment)	25
6.	Data Analysis Software (Spreadsheet/Statistical/Scripting Tools)	25
7.	Digital Multimeter	25
8.	ESD-safe Workbench with ESD Mat and Wrist Strap	25
9.	Sample IC Devices and Test Boards (Digital, Analog, Mixed-Signal)	25
10.	Breadboard and Electronic Component Kits	25
11.	Digital Storage Oscilloscope	13
12.	DC Regulated Power Supply	13
13.	Function/Signal Generator	13
14.	Logic Analyzer	13
15.	Sample Loadboard/DUT Board and Socket Demonstration Kits	05
16.	Probe Card and Test Fixture Illustration/Demonstration Set	02
17.	ATE Access or Remote/Partner ATE Facility (Advantest V93000 or equivalent) for Demonstration	01 set
18.	Sample Production Test Data Sets (STDF/CSV) with Analysis Templates	01 set
19.	Internet Connectivity with Network Switch and Router	01

S.N.	Major Training Equipment and Facilities	Required Facilities
20.	Central File/Storage Server	01
21.	First Aid and Fire Safety Equipment	01 set

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of the training course on Semiconductor Testing Engineering, the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.